

BERT QA (Extractive Question Answering) – Notes

1. How BERT QA Works

Input

Question + Context

Example:

Question:
Who is Apple CEO?

Context:
Apple CEO is Steve Jobs.

Tokenizer converts them into:

[CLS] Question [SEP] Context [SEP]

2. BERT Encoder

BERT reads **both question and context together**.

Every token gets a **contextual vector**.

Example:

Apple → 768 features
CEO → 768 features
Steve → 768 features
Jobs → 768 features

Output shape:

Batch × Tokens × Hidden_Size

1 × 15 × 768

3. QA Head

QA Head is only **one Linear Layer**.

Linear(768 → 2)

Weight matrix:

768 × 2

Two output neurons:

Neuron 1 → Start Score

Neuron 2 → End Score

Every token passes through the same QA head.

Token Vector (768)

↓

Start Logit

End Logit

Output:

15 × 2

Hugging Face splits it into

Start Logits → 15

End Logits → 15

4. Selecting the Answer

Normally

```
Start = argmax(start_logits)
```

```
End = argmax(end_logits)
```

If decoding uses Top-K:

- Take top K start positions.
- Take top K end positions.
- Form valid spans.
- Score each span.

Example:

```
Span Score
```

```
=
```

```
Start Logit
```

```
+
```

```
End Logit
```

Choose the highest valid span.

Training

Dataset contains

```
Question
```

```
Context
```

```
Start Label
```

```
End Label
```

Example

```
Question:
```

```
Who founded Apple?
```

```
Context:
```

```
Apple was founded by Steve Jobs.
```

Start = Steve

End = Jobs

Loss

Two independent losses are computed.

Start Loss

+

End Loss

Usually

Total Loss

=

$(\text{Start Loss} + \text{End Loss}) / 2$

If

Start Prediction

Wrong

↓

High Loss

If

End Prediction

Correct

↓

Low Loss

Backpropagation updates weights using both losses.

Important Understanding

The model DOES NOT compare spans during training.

It compares

Predicted Start

vs

True Start

and

Predicted End

vs

True End

independently.

What Do The Weights Learn?

Weights are NOT memorizing answers.

They learn statistical patterns.

Example training pattern

Who is CEO?

↓

Person Name

or

Who founded X?

↓

Founder Name

These patterns become encoded inside the weights.

During Testing

Question

Who is Apple CEO?

Context

Apple CEO is Mike Tyson.

The model sees

Question Pattern

+

Person Name

↓

Extract Person

because this resembles the training distribution.

The same learned weights are reused.

Weights never change during inference.

Generalization

Training

↓

Patterns

↓

Weights

↓

New Input

↓

Apply Same Learned Function

↓

Prediction

If the new input is similar to the training distribution, predictions are usually good.

If the new input is outside the learned distribution, performance degrades.

Limitation 1 — No Reasoning

Example

Apple CEO is NOT Mike Tyson.

BERT may answer

Mike Tyson

Reason:

BERT QA extracts spans.

It does not reason about negation.

Limitation 2 — No Fact Verification

Example

Apple CEO is Batman.

BERT may answer

Batman

It assumes the context is true.

Limitation 3 — Cannot Detect Contradictions

Example

Apple CEO is Steve.

Later...

Apple CEO is Mike.

BERT extracts the most likely span.

It does not resolve contradictions.

Limitation 4 — Cannot Explain

GPT can answer

The context suggests...

Therefore...

BERT only returns

Steve Jobs

No explanation.

Limitation 5 — Sensitive to Training Distribution

If training mostly contains

Who

Founder

CEO

President

then the learned weights become very good at extracting person names for these question patterns.

Unusual wording or unseen patterns may reduce accuracy.

Why Combine BERT + GPT?

BERT

↓

Fast Span Extraction

GPT

↓

Reasoning

Negation

Explanation

Natural Language Generation

Pipeline

Question

↓

Retriever / Database

↓

BERT QA

↓

Candidate Answer

↓

GPT

↓

Reasoned Final Answer

Final Mental Model

Training Data

↓

Statistical Patterns

↓

Weights Learn These Patterns

↓

Inference

↓

New Input Produces New Contextual Vectors

↓

Same Learned Weights Are Applied

↓

Prediction

The weights represent **learned statistical relationships**, not stored facts.